Harsha Vardhan

AI Engineer · Agentic AI · Voice AI Agents · Production LLM Systems

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Education

Indian Institute of Technology Bhubaneswar Jul 2021 – May 2025
B.Tech in Electronics and Communication Engineering (CGPA 7.75/10) *Bhubaneswar, India*

Experience

Imbesideyou Inc. Dec 2025 – Present
AI Engineer *Tokyo, Japan (Remote)*

- Built the Rust/Tauri desktop capture agent for an enterprise **process-mining platform** serving **NTT DATA** and **Toyota** (**¥15M+ monthly revenue**); cut frame capture from 2–8 s to ~83 ms.
- Led the capture contract feeding the OCR/LLM pipeline, replacing screenshot OCR with UI Automation, Office COM and browser signals (coverage 62.5% → 87%).
- Built a **multi-agent** automation-delivery system on **Claude** (17 subagents, **MCP server**, LLM-as-judge **evals**), growing a provenanced process record to ~3K facts.
- Designed the platform's identity layer: token-exchange service with audience-scoped JWTs and email-OTP **MFA**, and closed 5 cross-tenant access holes.
- Owned a **voice-AI SaaS** end to end for **DeCoA** and **Mitsubishi Motors** (**¥8M+ monthly revenue**): **Vapi** agents turning phone calls into reports, HR analytics and a compliance hotline.
- Tuned Japanese STT/TTS, built a call-concurrency limiter (0 errors at 200 concurrent requests) and capped per-tenant LLM spend via **Langfuse**.

ZS Associates Jul 2025 – Dec 2025
Software Engineer *Pune, India*

- Engineered **Databricks** and **AWS Redshift** pipelines for a **$2M** data-analytics program at healthcare client **Amgen**, contributing to **$45K in cost savings**.
- Built the program's client-side **React** interface, turning complex backend data into dashboards for business decision-makers.
- Built **LLM-powered analysis tools** in Python with **Langfuse** tracing and authentication, making automated data insights observable and reliable.

Projects

EchoChat | *Next.js, TypeScript, GetStream, ZegoCloud, Clerk*

- Built a real-time messaging platform with 1:1 and group chat, live notifications, video meetings and an integrated **AI chatbot**.

Smart Naka | *React Native, TypeScript*

- Built a mobile app that reads number plates in real time and cross-references a stolen-vehicle database; **Top 25 of 1,000+ teams**, Ethos Hackathon (IIT Guwahati).

Technical Skills

AI & LLM Systems: Claude (tool calling, subagents, prompt caching), MCP, OpenAI, AWS Bedrock, multi-agent systems, LLM-as-judge evals, Langfuse, embeddings, PM4Py

Voice AI: Vapi, Twilio/SIP, Soniox STT, Cartesia TTS, n8n

Languages: Python, TypeScript, Rust, SQL, JavaScript, C++

Backend & Cloud: FastAPI, PostgreSQL, Node.js, React, Firebase (Firestore, Cloud Functions, Cloud Run), AWS (CDK, ECS, Batch, Lambda, S3), Databricks, Docker, GitHub Actions

Achievements

- **JEE Advanced 2021:** All India Rank 3,348 among 1.4M+ candidates.
- **Inter-IIT Tech Meet 11.0 (IIT Kanpur):** Silver medal (Cloud Physician) and Bronze medal (Cognitive Garage).
- **Hackathons:** Top 25 of 1,000+ teams at Ethos (IIT Guwahati); 3rd of 1,500 participants at CodeRelay (IIT Bhubaneswar).
- **Club Head**, Software Development Club (Neuromancers), IIT Bhubaneswar: ran WinterCode Dash, a series of 4 weekend hackathons.